DSC 140B - Quiz 07

February 26, 2026

Name:

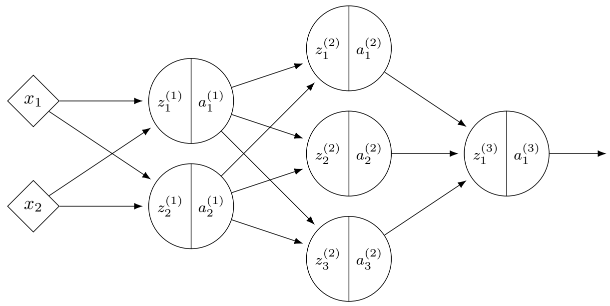
PID:

About the quizzes:

- Quizzes in DSC 140B are *optional* and graded pass/fail.
- A score of 70% or higher earns a “pass” and 1.5 credits toward your final grade.
- If you don’t pass, no credits are earned, but it doesn’t hurt your grade.
- You have 30 minutes to complete the quiz.
- At least one of the questions below will be on an exam (probably with slight changes, such as different numbers).
- Unfortunately, we can’t answer clarifying questions during the quiz. If you think a question has a bug or is unclear, please let us known in a private post on Campuswire after the quiz, and we’ll take it into account when grading.

Problem 1. (4 points)

Consider a neural network $H(\vec{x})$ shown below:



Assume that all hidden nodes use **ReLU activation** functions, that the output node uses a linear activation, and that there are no biases.

Let the weights of the second layer of the network be: $W^{(2)} = \begin{pmatrix} 3 & -1 & 2 \\ 1 & 4 & -2 \end{pmatrix}$.

In addition, suppose it is known that, when the network is evaluated at $\vec{x}$ with weight vector $\vec{w}$:

$$z_1^{(1)} = 4, \qquad z_2^{(1)} = -1$$

$$\partial H / \partial z_1^{(2)} = 2, \qquad \partial H / \partial z_2^{(2)} = -1, \qquad \partial H / \partial z_3^{(2)} = 3$$

a) What is $\partial H / \partial W_{11}^{(2)}$, as a number?

8

Solution: 8.

$$\frac{\partial H}{\partial W_{11}^{(2)}} = a_1^{(1)} \cdot \frac{\partial H}{\partial z_1^{(2)}} = \text{ReLU}(4) \cdot 2 = 4 \cdot 2 = 8.$$

This was similar to Practice Problem 142.

b) What is $\partial H / \partial W_{21}^{(2)}$, as a number?

0

Solution: 0.

$$\frac{\partial H}{\partial W_{21}^{(2)}} = a_2^{(1)} \cdot \frac{\partial H}{\partial z_1^{(2)}} = \text{ReLU}(-1) \cdot 2 = 0 \cdot 2 = 0.$$

Since $z_2^{(1)} = -1 < 0$, we have $a_2^{(1)} = \text{ReLU}(-1) = 0$.

This was similar to Practice Problem 142.

c) What is $\partial H / \partial a_1^{(1)}$, as a number?

13

Solution: 13.

$$\begin{aligned}\frac{\partial H}{\partial a_1^{(1)}} &= \sum_k W_{1k}^{(2)} \frac{\partial H}{\partial z_k^{(2)}} \\ &= 3(2) + (-1)(-1) + 2(3) \\ &= 6 + 1 + 6 = 13\end{aligned}$$

This was similar to Practice Problem 142.

d) What is $\partial H / \partial z_2^{(1)}$, as a number?

0

Solution: 0.

First, $\dfrac{\partial H}{\partial a_2^{(1)}} = \sum_k W_{2k}^{(2)} \dfrac{\partial H}{\partial z_k^{(2)}} = 1(2) + 4(-1) + (-2)(3) = -8.$

Then $\dfrac{\partial H}{\partial z_2^{(1)}} = \dfrac{\partial H}{\partial a_2^{(1)}} \cdot g'(z_2^{(1)}) = -8 \cdot g'(-1) = -8 \cdot 0 = 0.$

Since $z_2^{(1)} = -1 < 0$, the ReLU derivative is 0.

This was similar to Practice Problem 142.

Problem 2.

Consider a deep neural network $H(\vec{x})$ whose output layer uses a **sigmoid** activation function. For a particular input $\vec{x}$, the network's output is $H(\vec{x}) \approx 0$ (that is, it is approximately zero).

True or False: for this input, $\partial H / \partial W_{ij}^{(\ell)} \approx 0$ for all i, j, ℓ.

- ● True
- ○ False

> **Solution:** True.
>
> Since the output of the network is approximately zero, the input to the output node must be a large negative number. In this regime, the sigmoid is "saturated" and its plot is very flat, meaning that the derivative of the sigmoid is approximately zero at this point.
>
> Since every partial derivative $\partial H / \partial W_{ij}^{(\ell)}$ includes the factor $\sigma'(z^{(\text{out})})$ (by the chain rule through the output node), *all* partial derivatives are approximately zero.
>
> This was similar to Practice Problem 146.

Problem 3. (2 points)

Recall that the gradient of the square loss at a single data point $(\vec{x}, y)$ is:

$$\nabla_{\vec{w}} L = 2(\text{Aug}(\vec{x}) \cdot \vec{w} - y) \, \text{Aug}(\vec{x})$$

Suppose you are running **stochastic** gradient descent to minimize the empirical risk with respect to the square loss in order to train a function $H(x) = w_0 + w_1 x$ on the following data set:

x	y
2	3
1	-1
3	5
4	7
5	2

Suppose that the initial weight vector is $\vec{w} = (0, 0)^T$, the learning rate is $\eta = 1/2$, and the mini-batch size is 2. If the first mini-batch consists of the 2nd and 3rd data points, what will be the weight vector after one iteration of stochastic gradient descent?

Hint: the first entry should be 2.

$(2, 7)^T$

> **Solution:** $(2, 7)^T$.
>
> The mini-batch consists of the 2nd and 3rd data points: $(1, -1)$ and $(3, 5)$. We compute the gradient of the square loss at each of these points using $2(\text{Aug}(\vec{x}) \cdot \vec{w} - y) \, \text{Aug}(\vec{x})$.
>
> Since $\vec{w} = \vec{0}$, we have $\text{Aug}(\vec{x}) \cdot \vec{w} = 0$ for both points.
>
> Point $(1, -1)$: $2(0 - (-1))(1, 1)^T = (2, 2)^T$
>
> Point $(3, 5)$: $2(0 - 5)(1, 3)^T = (-10, -30)^T$

The gradient of the risk on the mini-batch is the average:

$$\frac{1}{2}\left(\begin{pmatrix}2\\2\end{pmatrix}+\begin{pmatrix}-10\\-30\end{pmatrix}\right)=\frac{1}{2}\begin{pmatrix}-8\\-28\end{pmatrix}=\begin{pmatrix}-4\\-14\end{pmatrix}$$

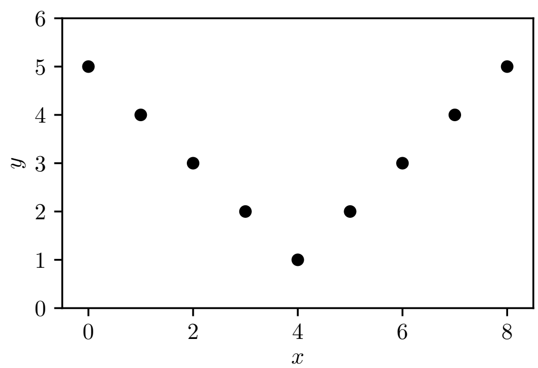
Applying the update rule $\vec{w}^{(1)} = \vec{w}^{(0)} - \eta \nabla R(\vec{w}^{(0)})$ with $\eta = 1/2$:

$$\vec{w}^{(1)} = \begin{pmatrix}0\\0\end{pmatrix} - \frac{1}{2}\begin{pmatrix}-4\\-14\end{pmatrix} = \begin{pmatrix}2\\7\end{pmatrix}$$

This was similar to Practice Problem 157.

Problem 4.

Suppose you want to train a neural network with a **single hidden layer** to predict y from x on the following one-dimensional regression data set:



True or False: if the hidden layer has sufficiently many nodes and uses **linear** activations (with a linear output node), the network can achieve zero empirical risk on this data set.

○ True

● False

Solution: False.

A neural network with linear activations, no matter how many hidden nodes it has, computes a linear (affine) function of its input. This is because the composition of linear functions is itself linear. The data shown follows a nonlinear pattern (a "V" shape), so no linear function can pass through all of the points, and the empirical risk cannot be zero. Nonlinear activations (such as ReLU or sigmoid) are required.

Problem 5.

Suppose you are training a neural network with two hidden layers of 64 nodes each, using ReLU activations.

True or False: it is typical to initialize all of the weights of this network to zero before training.

○ True

● False

Solution: False.

If all weights are initialized to zero, every hidden node in a given layer computes the same output, and during backpropagation, every node receives the same gradient update. As a result, the nodes never differentiate from one another — this is known as the *symmetry problem*. In practice, weights are initialized randomly (e.g., using small random values) to break this symmetry.

Problem 6.

Suppose you are training a deep neural network with 5 hidden layers, each using ReLU activations, to perform regression. You use stochastic gradient descent with a fixed learning rate to minimize the empirical risk with respect to the squared loss.

True or False: if trained for sufficiently many epochs, SGD is guaranteed to find a global minimum of the empirical risk.

○ True

● False

Solution: False.

The empirical risk function for a neural network is generally non-convex, meaning it can have many local minima and saddle points. Stochastic gradient descent has no guarantee of finding a global minimum in non-convex settings; it may converge to a local minimum or saddle point instead.